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Acid Ball Valve Assembly for Railroad Tank Car

Abstract

A ball valve assembly for a railroad tank car is disclosed herein. The ball valve assembly may comprise a ball valve removably coupled to an end of a fittings flange of a railroad tank car, a dual thread retainer assembly affixed to an end of the ball valve opposite the fittings flange, and an anti-rotation assembly attached to the fittings flange. The dual thread retainer assembly may comprise a retainer affixed to the end of the ball valve opposite the fittings flange and a cap removably affixed to an end of the retainer opposite the ball valve. The anti-rotation assembly may comprise a base affixed to the fittings flange and a fork secured to the base that is configured to prevent rotation of the ball valve and the dual thread retainer assembly with respect to the fittings flange.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/559,517filed Feb. 29, 2024, the content of which is hereby incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to railroad tank cars and, more particularly, to acid ball valves used on railroad tank cars.

BACKGROUND OF THE INVENTION

[0003] Railroad tank cars are frequently used to transport a variety of liquid or gaseous commodities, such as acid, fertilizer, polymer, crude oil, food grain products, and/or other goods or resources. Acid ball valves are frequently used on railroad tank cars because they can handle acid or corrosive fluids, and/or other resources. However, the threaded portion of acid ball valves often suffer wear and tear during installation and/or maintenance operations due, for example, to over-torquing when installing or tightening the valve to a tank car or installing a plug into the valve. This wear and tear and/or over-torquing can cause stripping to the threaded portion or stretch the threaded portion beyond its limits. This can result in loose plugs, valves, or loading equipment which may cause leaks from the tank car when there is positive pressure in the tank car. Such leaks must be detected and repaired before the tank car continues its route. In addition, due to vibration of the valve during transportation, there may be a risk of potential rotation or twisting of an installed ball valve and/or additional components coupled to a component of a ball valve assembly. To ensure acid ball valves are in proper working order, the valves also need to be inspected to ensure there are not any leaks. Any acid ball valve assembly that can reduce the aforementioned risks would represent an improvement over conventional acid ball valves used on railroad tank cars.

SUMMARY OF THE INVENTION

[0004] Aspects of this disclosure relate to a ball valve assembly for a railroad tank car. In various embodiments, the ball valve assembly comprises a ball valve removably coupled to an end of a fittings flange of a railroad tank car, a dual thread retainer assembly affixed to an end of the ball valve opposite the fittings flange, and an anti-rotation assembly securely attached to the fittings flange to prevent rotation of the ball valve and the dual thread retainer assembly with respect to the fittings flange. In various embodiments, the dual thread retainer assembly comprises a retainer affixed to the end of the ball valve opposite the fittings flange and a cap removably affixed to an end of the retainer opposite the ball valve. In various embodiments, the cap is removably affixed to the retainer via an inner threaded portion of the cap and a first outer threaded portion of the retainer. In some embodiments, the first outer threaded portion includes one or more channels to allow leaks to be detected prior to removing the cap from the retainer.

[0005] In various embodiments, the retainer further comprises a second outer threaded portion via which the retainer is affixed to the end of the ball valve opposite the fittings flange. In some embodiments, the dual thread retainer assembly further comprises an O-ring at the interface between the cap and the retainer that is configured to allow the cap to be sufficiently tightened (e.g., to form a seal) when the cap is affixed to the retainer. In some embodiments, the O-ring is configured to be retained within the cap when the cap is removed from the retainer. In some embodiments, the end of the retainer opposite the ball valve includes a tapered outer edge. In such embodiments, the tapered outer edge may be radiused for fitment of the O-ring. For example, the tapered outer edge may be sized such that it is configured to stretch the O-ring to a particular size

and align the O-ring with respect to the retainer when the cap is affixed to the retainer.
[0006] According to another aspect of this disclosure, an anti-rotation assembly for use with a ball valve assembly affixed to a railroad tank car is described herein. The anti-rotation assembly may include a base affixed to the fittings flange and a fork secured to the base that is configured to prevent rotation of the ball valve with respect to the fittings flange. In various embodiments, the base is welded to an exposed surface of the fittings flange. In various embodiments, an opening in the fork and an opening in the base, when aligned, are each configured to receive a single fastener configured to secure the fork to the base. In some embodiments, the opening in the fork may comprise a slot configured to allow a position of the fork to be adjusted relative to the base. In various embodiments, the fork includes a portion that partially surrounds the ball valve and is configured to prevent rotation of the ball valve with respect to the fittings flange. In some embodiments, such a portion of the fork comprises a hexagonal grabbing profile configured to provide increased surface contact area with the ball valve.
[0007] According to another aspect of the invention, a method for installing a ball valve assembly at a fittings flange of a railroad tank car is disclosed herein. In various embodiments, the method comprises removably coupling a threaded nipple to an end of a fittings flange of a railroad tank car. In various embodiments, the method further comprises coupling a ball valve to the fittings flange via the threaded nipple. In various embodiments, the method further comprises securely attaching a base of an anti-rotation assembly to the fittings flange. In various embodiments, the method further comprises fastening a fork of the anti-rotation assembly to the base, wherein the fork is configured to prevent rotation of the ball valve with respect to the fittings flange. In some embodiments, securely attaching the base of the anti-rotation assembly to the fittings flange includes marking a position that provides a gap between the fork and a termination point of an interfacing region of the ball valve. In some embodiments, the method further comprises affixing a dual thread retainer assembly to an end of the ball valve opposite the fittings flange, the dual thread retainer assembly comprising a retainer affixed to the end of the ball valve opposite the fittings flange and a cap removably affixed to an end of the retainer opposite the ball valve. In some embodiments, the method further comprises removably affixing a cap to the retainer via an inner threaded portion of the cap and a first outer threaded portion of the retainer, wherein the retainer further comprises a second outer threaded portion, and wherein the retainer is affixed to the end of the ball valve opposite the fittings flange via the second outer threaded portion. In some embodiments, the first outer threaded portion includes one or more channels to allow leaks to be detected prior to removing the cap from the retainer.
[0008] These and other objects, features, and characteristics of the invention disclosed herein will become more apparent upon consideration of the following description and the appended claims with reference to the accompanying drawings, all of which form a part of this specification, wherein like reference numerals designate corresponding parts in the various figures. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only and are not intended as a definition of the limits of the invention. As used in the specification and in the claims, the singular form of “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] The present invention is illustrated by way of example and not limited in the accompanying figures in which like reference numerals indicate similar elements and in which:
[0010] FIG. **1** depicts a perspective view of an example acid ball valve assembly for a railroad tank car, according to one or more aspects described herein;

[0011] FIGS. **2**A-B depict cross-sectional views of an example ball valve assembly, according to one or more aspects described herein;
[0012] FIGS. **3**A-B depict perspective views of an example ball valve assembly with a valve handle in an open and closed position, according to one or more aspects described herein;
[0013] FIGS. **4**A-D depict various views of an example dual thread retainer assembly for a ball valve assembly, according to one or more aspects described herein;
[0014] FIGS. **5**A-C depict various additional views of an example dual thread retainer assembly for a ball valve assembly, according to one or more aspects described herein;
[0015] FIGS. **6**A-E depict various views of an example ball valve assembly with an anti-rotation assembly, according to one or more aspects described herein;
[0016] FIGS. **7**A-B depict various views of an example ball valve assembly with an anti-rotation assembly, according to one or more aspects described herein;
[0017] FIGS. **8**A-C depict various views of an example ball valve assembly with a safety chain, according to one or more aspects described herein;
[0018] FIGS. **9**A-B depict perspective views of alternative embodiments of one or more components of a ball valve assembly, according to one or more aspects described herein;
[0019] FIGS. **10**A-E provide a pictorial representation of an example process for installing a ball valve assembly to a fittings flange, according to one or more aspects described herein;
[0020] FIG. **11** depicts a flow diagram of an example method for installing an example ball valve assembly mounted to a railroad tank car, according to one or more aspects described herein.
[0021] These drawings are provided for purposes of illustration only and merely depict typical or example embodiments. These drawings are provided to facilitate the reader's understanding and shall not be considered limiting of the breadth, scope, or applicability of the disclosure. For clarity and ease of illustration, these drawings are not necessarily drawn to scale.

DETAILED DESCRIPTION

[0022] In the following description of various examples of the invention, reference is made to the accompanying drawings, which form a part hereof, and in which are shown by way of illustration various example structures, systems, and steps in which aspects of the invention may be practiced. These aspects are indicative, however, of but a few of the various ways in which the principles of the invention may be employed and the present invention is intended to include all such aspects and their equivalents. It is to be understood that other specific arrangements of parts, structures, example devices, systems, and steps may be utilized, and structural and functional modifications may be made without departing from the scope of the present invention. Also, while the terms “top,” “bottom,” “front,” “back,” “side,” and the like may be used in this specification to describe various example features and elements of the invention, these terms are used herein as a matter of convenience, e.g., based on the example orientations shown in the figures. Nothing in this specification should be construed as requiring a specific three-dimensional orientation of structures in order to fall within the scope of this invention.
[0023] The invention described herein relates to a ball valve assembly for a railroad tank car configured to reduce the risk of potential leaks due to the wear and tear that may occur on a threaded portion of a ball valve. Another aspect of the invention described herein relates to a ball valve assembly for a railroad tank car that reduces risk of potential rotation or twist of the ball valve and/or one or more flow control components that may surround an outer portion of the ball valve.
[0024] For convenience, the ball valve assembly and methods of installing a ball valve described herein may comprise or be suitable for use with an acid ball valve affixed (or configured to be affixed) to a railroad tank car. However, it is to be understood that the ball valve assembly and methods of installing the same described herein may be equally applicable in other applications without departing from the scope of the invention. For example, the ball valve described herein may also be utilized in other applications, such as in other applications involving tanks or other

storage containers to which a ball valve may be installed.
[0025] FIG. **1** depicts a perspective view of an acid ball valve assembly **100** for a railroad tank car, according to one or more aspects described herein. In various embodiments, ball valve assembly **100** may include a ball valve **120**, dual thread retainer assembly **200** affixed to an end of ball valve **120**, an anti-rotation assembly **300**, and/or one or more other components. In various embodiments, ball valve **120** may be removably coupled to an end of a fittings flange **55** via a threaded nipple **60**. The anti-rotation assembly **300** may be securely attached to the fittings flange **55** to prevent rotation of the ball valve **120** and dual thread retainer assembly **200** with respect to the fittings flange **55** and/or one or more other components. In various embodiments, ball valve assembly **100** may be configured to reduce the risk of potential leaks due to wear and tear that may occur on threaded portions of the ball valve.
[0026] FIGS. **2**A-B depict cross-sectional views of ball valve assembly **100**, according to one or more aspects described herein. In various embodiments, ball valve assembly **100** may include a ball valve **120**, dual thread retainer assembly **200** affixed to an end of the ball valve **120**, an anti-rotation assembly **300** (not shown in FIGS. **2**A-B), and/or one or more other components. In various embodiments, ball valve **120** may include a valve body **122**, valve stem **140**, a valve handle **160**, and/or one or more other components. In some embodiments, valve handle **160** may be rotatably movable with respect to the valve stem **140** between an open position and a closed position. In various embodiments, valve body **122** of ball valve assembly **100** may define an interior cavity of the ball valve **120** that is revealed by rotating a valve handle **160** from a closed position **162** to an open position **164**, for example, as depicted in FIGS. **3**A-B.
[0027] Indeed, FIGS. **3**A-B depict various views of ball valve **120** with a valve handle **160**, according to one or more aspects described herein. In various embodiments, valve handle **160** of ball valve assembly **100** may be slidably rotated from a closed position **162** to an open position **164** to control flow path of ball valve assembly **100**. For example, in some implementations, ball valve **120** may be a quarter turn valve, in which the valve handle **160** turns only **90** degrees from a fully closed position **162** to a fully open position **164** of the valve.
[0028] Returning to FIGS. **2**A-B, in various embodiments, valve stem **140** may be mated to and received within an opening of valve body **122**. In various embodiments, valve stem **140** may comprise a stem base **142** and a stem body **144** that extends outwardly to receive and engage with valve handle **160** of ball valve assembly **100**. For example, as depicted in FIG. **2**A, stem body **144** may be configured to extend away from stem base **142** along an axis perpendicular to a longitudinal axis A-A′ along which ball valve assembly **100** is mounted. In various embodiments, valve stem **140** may be configured to receive nuts, washers, and/or other types of fasteners used to secure valve handle **160** to ball valve assembly **100**. For example, valve stem **140** may be configured to receive nuts, washers, and/or other types of fasteners used to removably attach valve handle **160** to the valve stem **140** of ball valve **120**.
[0029] In various embodiments, the valve body **122** may include a first valve threaded portion and a second valve threaded portion along a longitudinal axis A-A′. In some embodiments, ball valve assembly **100** may be coupled to an end of a fittings flange **55** via a threaded nipple **60** (as depicted in FIG. **1**) at the first valve threaded portion using one or more techniques described herein and/or using any other now known or future developed fastening technique. For example, a tapered threaded configuration may be utilized at the fittings flange **55** on a top or side of a railroad tank car to be coupled with the valve body **122** of ball valve assembly **100**. While a tapered threaded portion is described herein, it should be understood that any appropriate modification of the first and second threaded portion may be contemplated and is within the scope of the invention described herein.
[0030] In various embodiments, ball valve assembly **100** described herein may include a dual thread retainer assembly **200** comprising a retainer **240** and a removable cap **260**. In various embodiments, retainer **240** may be affixed to an end of the ball valve **120**. For example, retainer

240 may be welded (or securely attached) to valve body **122** once it is threaded in such that it forms a functional part of valve body **122** (and ball valve assembly **100**). In various embodiments, retainer **240** of ball valve assembly **100** may be substantially cylindrical with a retainer base at the base to mate with the ball valve **120**. As described herein, in various embodiments, retainer **240** may comprise a retainer body **242** having at least side walls **244**, a retainer base **246**, and/or one or more other components. In various embodiments, retainer **240** may be mated to and received within an opening of valve body **122**. As depicted in FIG. **2**A, in various embodiments, retainer **240** may comprise the retainer base **246** and the retainer body **242** that extends axially towards cap **260**. For example, retainer body **242** may be configured to extend away from retainer base **246** along a longitudinal axis perpendicular to a mounting surface to which ball valve assembly **100** is mounted.
[0031] In some embodiments, as depicted in FIG. **2**A, cap **260** may include an O-ring **270** configured to seal interface between cap **260** and retainer **240**. In some embodiments, O-ring **270** may be constructed of an elastomer material that can deform to fill surface imperfections in the groove **268** of cap **260** and the interior of retainer **240** (as depicted in FIG. **5**B), thereby enabling tighter sealing of ball valve assembly **100**. The O-ring **270** may allow metal to metal contact of cap **260** and retainer **240**, which enables cap **260** to be torqued sufficiently tight (regardless of cap torque value) without over-compressing O-ring **270**. In some embodiments, O-ring **270** (and cap **260**) may be configured such that O-ring **270** is retained within cap **260** when cap **260** is removed from retainer **240**. In alternative embodiments, and as depicted in FIG. **2**B and/or FIG. **5**C, cap **260** may include a gasket ring **280** configured to seal interface between cap **260** and retainer **240**.
[0032] In various embodiments, retainer body **242** of retainer **240** may include a first outer threaded portion **252**, a second outer threaded portion **254**, and an inner threaded portion **256**. In various embodiments, a cap **260** may be removably engaged to a top side of retainer **240**. The cap **260** may include an inner threaded portion **262** corresponding to a first outer threaded portion **252**. In various embodiments, dual thread retainer assembly **200** may include a removable cap **260** coupled to an end of a retainer **240**, which itself is affixed to an end of the ball valve **120**. FIGS. **4**A-D depict various views of an example dual thread retainer assembly **200** for ball valve assembly **100**, according to one or more aspects described herein. For example, FIG. **4**A depicts a perspective view of cap **260** of ball valve assembly **100**, FIG. **4**B depicts a cross-sectional view of the cap **260** as depicted in FIG. **4**A along a line B-B′, FIG. **4**C depicts a perspective view of retainer **240** of ball valve assembly **100**, and FIG. **4**D depicts a cross-sectional view of the retainer **240** as depicted in FIG. **4**C along a line C-C′. In various embodiments, cap **260** may include one or more portions with a vertically varying profile **264**. For example, the vertically varying profile may be present in outer portion of cap **260** extending along the line B-B′ to provide ease of use and further engagement between cap **260** and retainer **240** with minimum torque. In some embodiments, the vertically varying profile **264** of cap **260** may include (or form) one or more grooves. In some embodiments, grooves in (or on) cap **260** may used, for example, to retain a safety chain, such as safety chain **390** depicted in and described herein with respect to FIGS. **8**A-C.
[0033] In some embodiments, as depicted, for example, in FIGS. **4**C-D, retainer **240** may include a tapered outer edge **258** to receive an O-ring (such as O-ring **270** depicted in FIG. **2**A and described herein). In other words, retainer **240** may include a chamfer (or tapered edge) at a top portion of the retainer **240** configured to enable an easy entry and/or exit of an O-ring to reduce risk of any potential damage during compression. In some embodiments, tapered outer edge **258** may be configured to provide room for an O-ring to slide into position while ensuring the O-ring is not damaged during compression. For example, tapered outer edge **258** may be radiused for fitment of an O-ring. In various embodiments, tapered outer edge **258** may be sized such that it is configured to stretch the O-ring to a particular size and align the O-ring with respect to retainer **240** when cap **260** is affixed to retainer **240**. For example, tapered outer edge **258** may be configured to help O-ring **270** slide into position when cap **260** is screwed on. In such embodiments, the taper on tapered outer edge **258** may be configured to allow for optimized compression on O-ring **270** and its

tapered and radius shape ensures O-ring **270** can stretch into position without being pinched or cut.

[0034] In various embodiments, cap **260** may be removably engaged to a top side of retainer **240**. As depicted in FIGS. **4**B-C, cap **260** may include an inner threaded portion **262** corresponding to a first outer threaded portion **252** of retainer **240**. For example, FIG. **2**A depicts an embodiment of ball valve assembly **100** in which cap **260** is removably engaged to retainer **240** by dual threaded portions (or similar mechanism) including first outer threaded portion **252** and inner threaded portion **262** to reduce potential wear and tear that may occur on threaded portions of the ball valve such as internal threaded portion **256**. In some embodiments, cap **260** may be made with any appropriate type of inner threaded portion **262** with a diameter, height, and/or shape of threaded portion selected to match diameter, height, and/or shape of corresponding first outer threaded portion **252** of retainer **240**. As depicted in FIGS. **5**A-B, cap **260** may be rotatably inserted over the corresponding (i.e., first outer threaded portion **252**) portion of retainer **240**.

[0035] FIGS. **5**A-B depict various additional views of dual thread retainer assembly **200** for ball valve assembly **100**, according to one or more aspects described herein. As described herein, ball valve assembly **100** may be configured to allow for improved control of the ball valve during installation on a railroad tank car. In various embodiments, ball valve assembly **100** may be configured to allow leaks to be detected without requiring cap **260** to be removed from the fittings flange **55** or requiring disassembling any components on the valve or the railroad tank car itself. For example in some embodiments, one or more channels **248** may be integrated in first outer threaded portion **252** of retainer **240** (as depicted in FIG. **5**A) to allow leaks to be detected prior to removing the cap **260** from the retainer **240**.

[0036] In various embodiments, the one or more channels **248** may extend axially (i.e., parallel to a longitudinal axis of ball valve assembly **100** along a straight path). For example, the one or more channels **248** may extend along a line C-C′, as depicted in FIG. **4**C. In other embodiments, the one or more channels **248** may extend axially along a curved path. In some embodiments, the one or more channels **248** may comprise a rectangular opening along the straight path. In other embodiments, the one or more channels **248** may comprise a circular opening (e.g., when viewed from a vertical cross-sectional view) along the straight path. While one or more channels is described herein, it should be understood that any appropriate modification of the one or more channels **248** may be contemplated and is within the scope of the invention described herein. In various embodiments, and as depicted in FIG. **5**A, a width of one end of one or more channels **248** may be the same width of the other end of one or more channels **248** along a line B-B′. In other embodiments, a width of one end of one or more channels **248** may include a different width of the other end of one or more channels **248** along a line C-C′. In some embodiments, the one or more channels **248** may include a channel profile to enhance the detection of the leak. In various embodiments, the one or more channels **248** may include any suitable number of channels. For example, as depicted in FIG. **5**A, retainer **240** may include a single channel. However, retainer **240** may include two or more channels. In such embodiments, the one or more channels **248** may be equally spaced apart circumferentially around the first outer threaded portion **252** of retainer **240**. In other embodiments, the one or more channels **248** may be generally spaced apart circumferentially around the first outer threaded portion **252** of retainer **240**.

[0037] In various embodiments, cap **260** may include a groove **268**, notch, or other type of depression configured to receive an O-ring **270** configured to seal off ball valve **120** at the dual thread retainer assembly **200** (i.e., between cap **260** and retainer **240**) when ball valve **120** is in a closed position (as depicted, for example, in FIG. **5**B). For example, cap **260** and retainer **240** may, together, form a groove **268** that makes compression effectively identical every time cap **260** is installed regardless of cap installation torque. In various embodiments, O-ring **270** may form a seal when ball valve **120** is in a closed position. In some embodiments, O-ring **270** may be constructed of an elastomer material that can deform to fill surface imperfections in the groove **268** of cap **260** and the interior of retainer **240**, thereby enabling tighter sealing of ball valve assembly **100**. In alternative embodiments, as depicted in FIG. 5C, cap **260** may include a gasket ring **280**. For example, gasket ring **280** may comprise a quad ring or O-ring configured to form a seal when ball valve **120** is in a closed position. In some embodiments, gasket ring **280** may be constructed of an elastomer material that can deform to fill surface imperfections in the groove **268** of cap **260** and the interior of retainer **240**, thereby enabling tighter sealing of ball valve assembly **100**.

[0038] According to another aspect of the invention, ball valve assembly **100** described herein may include an anti-rotation assembly **300**. FIGS. **6**A-E depict various views of ball valve assembly **100** with an anti-rotation assembly **300**, according to one or more aspects described herein. In various embodiments, the anti-rotation assembly **300** may include an anti-rotation base (or stop rod) **302**, an anti-rotation fork **304**, one or more anti-rotation fasteners **306** passing through one or more anti-rotation openings **308** and **310** of the anti-rotation base **302** and the anti-rotation fork **304**, respectively. FIGS. **6**B-C depict cross-sectional views of anti-rotation fork **304** as depicted in FIG. **6**A along a line E-E′, and FIGS. **6**D-E depict cross-sectional views of anti-rotation base **302** as depicted in FIG. **6**A. In various embodiments, anti-rotation assembly **300** may be securely attached to fittings flange using one or more coupling elements and one or more techniques described herein and/or using any other now known or future developed fastening technique. For example, in some embodiments, anti-rotation assembly **300** (as depicted, for example, in FIGS. **6**A-E) may be welded to an exposed surface of fittings flange **55** (or the tank car) and removably attached to the ball valve assembly **100** via anti-rotation fork **304** and/or one or more anti-rotation fasteners **306**. In other embodiments, anti-rotation assembly **300** may be positioned surrounding the fittings flange **55** and attached to anywhere outside of fittings flange **55** by any appropriate fastening configuration.

[0039] In various embodiments, ball valve **120** and/or dual thread retainer assembly **200** would function the same way with or without the anti-rotation assembly **300**. However, anti-rotation assembly **300** may greatly reduce the risk of potential rotation or twist of ball valve **120** and/or dual thread retainer assembly **200** relative to fittings flange **55** and one or more flow control components that may surround outer portion of the ball valve assembly **100**.

[0040] FIGS. **7**A-B depict various views of ball valve assembly **100** with anti-rotation assembly **300**, according to one or more aspects described herein. In various embodiments, anti-rotation assembly **300** may include an anti-rotation fork **304** with a slot **310** to allow the position of one or more anti-rotation fasteners **306** to be adjusted and to further provide benefits of rotational freedom when installing or tightening the ball valve for installation or transportation. For example, in some embodiments, one or more anti-rotation fasteners **306** (e.g., bolts) may be used to attach the anti-rotation fork **304** to anti-rotation base **302** via a slot **310**, and may be adjusted in a range of an angle 0 from about 5 degrees (i.e., −5° to 5°)to about 15 degrees (i.e., −15° to 15°) relative to the valve body **122**. In a particular embodiment, the range of an angle 0 may be about 10 degrees (i.e., −10° to 10°). In various embodiments, and as depicted in FIG. **7**B, anti-rotation fork **304** may be made with any appropriate shape of grabbing profile **312** with a size and/or shape of grabbing profile selected to match size and/or shape of corresponding (i.e., grabbing) portion of ball valve **120**. For example, a half hexagonal grabbing profile may provide increased surface contact area with ball valve **120** than a traditional square fork. In various embodiments, the size and/or shape of the grabbing profile **312** of anti-rotation fork **304** may correspond to an outer edge of ball valve **120** and/or another component of ball valve assembly **100** such that a portion of anti-rotation fork **304** partially surrounds and prevents rotation of ball valve **120** (or another component of ball valve assembly **100**) with respect to the fittings flange **55**.

[0041] FIGS. **8**A-C depict various views of ball valve assembly **100** with a safety chain **390**, according to one or more aspects described herein. As described herein, in various embodiments, dual thread retainer assembly **200** of ball valve assembly **100** may include a removable cap **260**. In such embodiments, ball valve assembly **100** may include (or interface with) a safety chain **390** configured to be attached to removable cap **260** and any of the one or more other components of ball valve assembly **100** described herein or the tank car itself, such that safety chain **390** may secure removable cap **260** when removable cap **260** is unscrewed or otherwise detached from retainer **240**. In other words, safety chain **390** may serve as a tether to ensure removable cap **260** is not lost when removable cap **260** is removed from retainer **240**.

[0042] FIGS. **9**A-B depict perspective views of alternative embodiments of one or more components of ball valve assembly **100**, according to one or more aspects described herein. FIG. **9**B depicts a cross-sectional view of the ball valve **120** as depicted in FIG. **9**A along a line F-F′. In various embodiments, ball valve assembly **100** may include a ball valve **120** with a flanged connection **190** and a dual thread retainer assembly **200**, and/or one or more other components. For example, as depicted in FIGS. **9**A-B, the ball valve **120** may comprise a valve body **122** that comprises a flanged portion **190** at an end couple to fittings flange. In such embodiments, anti-rotation assembly **300** may not be needed because the flanged connection **190** may eliminate the risk of potential rotation or twist of ball valve **120** and/or dual thread retainer assembly **200** relative to fittings flange.

[0043] FIGS. **10**A-E provide a pictorial representation of an example process for installing a ball valve assembly **100** to fittings flange **55**, and FIG. **11** illustrates an example of a process **1100** for installing a ball valve assembly **100** to a fittings flange **55**, according to one or more aspects described herein. The operations of process **1100** presented below are intended to be illustrative and, as such, should not be viewed as limiting. In some implementations, process **1100** may be accomplished with one or more additional operations not described, and/or without one or more of the operations discussed. In some implementations, two or more of the operations of process **1100** may occur substantially simultaneously. The described operations may be accomplished using some or all of the components described in detail above with respect to ball valve assembly **100**.

[0044] In an operation **1102**, process **1100** may include removably coupling a threaded nipple **60** to an end of a fittings flange **55**, as depicted in FIG. **10**A. For example, the threaded nipple **60** may be installed and secured to the fittings flange **55**, as described herein. In an operation **1104**, process **1100** may include inserting a ball valve **120** engaged with dual thread retainer assembly **200** onto the threaded nipple **60** that is coupled to the end of a fittings flange, as depicted in FIG. **10**B. In an operation **1106**, process **1100** may include positioning anti-rotation assembly **300** (i.e., anti-rotation base **302** and the anti-rotation fork **304**) and mark a stop rod position that provides a gap between the end of the anti-rotation fork **304** closer to ball valve **120** and a termination point of an interfacing region of the valve body **122**, as depicted in FIG. **10**C. The gap allows room for future adjustments. In some embodiments, the gap may be in a range of a distance d from about 1/16 inch to about ¼ inch relative to the end of the anti-rotation fork **304** closer to ball valve **120**. In particular embodiments, the gap may be about ⅛ inch relative to the end of the anti-rotation fork **304** closer to ball valve **120**. In an operation **1108**, process **1100** may include securely attaching the anti-rotation base **302** to fittings flange **55**, for example, as depicted in FIG. **10**D, using one or more techniques described herein and/or using any other now known or future developed coupling technique. FIG. **10**D depicts a perspective view of an example anti-rotation base **302** without anti-rotation fork **304** that fits and surrounds an outer edge of fittings flange **55**. As depicted in FIG. **10**D, in some embodiments, anti-rotation assembly **300** may be welded to an exposed surface of fittings flange **55**. In an operation **1110**, process **1100** may include applying one or more anti-rotation fasteners **306** (e.g., bolts) to a position at which the one or more anti-rotation fasteners **306** (e.g., bolts) may be used to attach anti-rotation fork **304** to anti-rotation base **302** as depicted, for example, in FIG. **10**E, thereby completing an installation of the ball valve assembly **100** onto the fittings flange **55**. For example, as depicted in FIG. **10**E, anti-rotation fork **304** may then be slidably inserted over the ball valve **120** and attached to the anti-rotation base **302** (depicted, for example, in FIGS. **6**A-E) via one or more anti-rotation fasteners **306**.

[0045] It is to be understood that the invention is not limited in its application to the details of construction and the arrangement of the components set forth herein. The invention is capable of other embodiments and of being practiced or being carried out in various ways. Variations and modifications of the foregoing are within the scope of the present invention. It should be understood that the invention disclosed and defined herein extends to all alternative combinations of two or more of the individual features mentioned or evident from the text and/or drawings. All of these different combinations constitute various alternative aspects of the present invention. The embodiments described herein explain the best modes known for practicing the invention and will enable others skilled in the art to utilize the invention.

[0046] While the preferred embodiments of the invention have been shown and described, it will be apparent to those skilled in the art that changes and modifications may be made therein without departing from the spirit of the invention, the scope of which is defined by this description.

[0047] Reference in this specification to “one implementation”, “an implementation”, “some implementations”, “various implementations”, “certain implementations”, “other implementations”, “one series of implementations”, or the like means that a particular feature, design, structure, or characteristic described in connection with the implementation is included in at least one implementation of the disclosure. The appearances of, for example, the phrase “in one implementation” or “in an implementation” in various places in the specification are not necessarily all referring to the same implementation, nor are separate or alternative implementations mutually exclusive of other implementations. Moreover, whether or not there is express reference to an “implementation” or the like, various features are described, which may be variously combined and included in some implementations, but also variously omitted in other implementations. Similarly, various features are described that may be preferences or requirements for some implementations, but not other implementations.

[0048] The language used herein has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. Other implementations, uses and advantages of the invention will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. The specification should be considered exemplary only, and the scope of the invention is accordingly intended to be limited only by the following claims.

Claims

1. A ball valve assembly for a railroad tank car comprising: a ball valve removably coupled to an end of a fittings flange of a railroad tank car; a dual thread retainer assembly affixed to an end of the ball valve opposite the fittings flange, the dual thread retainer assembly comprising: a retainer affixed to the end of the ball valve opposite the fittings flange; and a cap removably affixed to an end of the retainer opposite the ball valve; and an anti-rotation assembly securely attached to the fittings flange to prevent rotation of the ball valve and the dual thread retainer assembly with respect to the fittings flange.

2. The ball valve assembly of claim 1, wherein the cap is removably affixed to the retainer via an inner threaded portion of the cap and a first outer threaded portion of the retainer.

3. The ball valve assembly of claim 2, wherein the first outer threaded portion includes one or more channels to allow leaks to be detected prior to removing the cap from the retainer.

4. The ball valve assembly of claim 2, wherein the retainer further comprises a second outer threaded portion, and wherein the retainer is affixed to the end of the ball valve opposite the fittings flange via the second outer threaded portion.

5. The ball valve assembly of claim 1, wherein the dual thread retainer assembly further comprises an O-ring at the interface between the cap and the retainer to allow the cap to be sufficiently tightened when the cap is affixed to the retainer.

6. The ball valve assembly of claim 5, wherein the O-ring is retained within the cap when the cap is removed from the retainer.

7. The ball valve assembly of claim 5, wherein the end of the retainer opposite the ball valve includes a tapered outer edge.

8. The ball valve assembly of claim 7, wherein the tapered outer edge is radiused for fitment of the O-ring.

9. The ball valve assembly of claim 7, wherein the tapered outer edge is configured to stretch the O-ring to a particular size and align the O-ring with respect to the retainer when the cap is affixed to the retainer.

10. The ball valve assembly of claim 1, wherein the anti-rotation assembly comprises a base affixed to the fittings flange and a fork secured to the base, wherein the fork is configured to prevent rotation of the ball valve and the dual thread retainer assembly with respect to the fittings flange.

11. The ball valve assembly of claim 10, wherein the base is welded to an exposed surface of the fittings flange.

12. The ball valve assembly of claim 10, wherein, when aligned, an opening in the fork and an opening in the base are each configured to receive a single fastener configured to secure the fork to the base.

13. The ball valve assembly of claim 10, wherein the opening in the fork comprises a slot configured to allow a position of the fork to be adjusted relative to the base.

14. The ball valve assembly of claim 10, wherein a portion of the fork partially surrounds and prevents rotation of the ball valve with respect to the fittings flange.

15. The ball valve assembly of claim 10, wherein the portion of the fork comprises a hexagonal grabbing profile configured to provide increased surface contact area with the ball valve.

16. A method for installing a ball valve assembly at a fittings flange of a railroad tank car, the method comprising: removably coupling a threaded nipple to an end of a fittings flange of a railroad tank car; coupling a ball valve to the fittings flange via the threaded nipple; securely attaching a base of an anti-rotation assembly to the fittings flange; and fastening a fork of the anti-rotation assembly to the base, wherein the fork is configured to prevent rotation of the ball valve with respect to the fittings flange.

17. The method of claim 16, wherein securely attaching the base of the anti-rotation assembly to the fittings flange includes marking a position that provides a gap between the fork and a termination point of an interfacing region of the ball valve.

18. The method of claim 16, the method further comprising: affixing a dual thread retainer assembly to an end of the ball valve opposite the fittings flange, the dual thread retainer assembly comprising a retainer affixed to the end of the ball valve opposite the fittings flange and a cap removably affixed to an end of the retainer opposite the ball valve.

19. The method of claim 18, the method further comprising removably affixing a cap to the retainer via an inner threaded portion of the cap and a first outer threaded portion of the retainer, wherein the retainer further comprises a second outer threaded portion, and wherein the retainer is affixed to the end of the ball valve opposite the fittings flange via the second outer threaded portion.

20. The method of claim 19, wherein the first outer threaded portion includes one or more channels to allow leaks to be detected prior to removing the cap from the retainer.